

**Vanishing criteria, coherence conditions for the
sheaves $\underline{\mathrm{Ext}}_Y^i(F, G)$
SGA 2, Exposé VII**

English draft

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About this draft

This is an unofficial English translation of Exposé VII, “Vanishing criteria, coherence of $\underline{\mathrm{Ext}}_Y^i(F, G)$ ”, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ VII

Vanishing criteria, coherence conditions for the sheaves

$$\underline{\mathrm{Ext}}_Y^i(F, G)$$

1. Study for $i < n$

Let us prove a lemma:

LEMMA 1.1. *Let X be a locally noetherian prescheme, Y a closed subset of X and G a quasi-coherent $\mathcal{O}_X$ -Module. Suppose that for every coherent $\mathcal{O}_X$ -Module F of support contained in Y , one has:*

$$\underline{\mathrm{Ext}}^{n-1}(F, G) = 0.$$

Then for every coherent $\mathcal{O}_X$ -Module F and every closed subset Z of X such that $Y \supset \mathrm{Supp} F \cap Z$, one has

$$\underline{\mathrm{Ext}}_Z^n(F, G) \approx \underline{\mathrm{Hom}}(F, \underline{H}_Y^n(G)).$$

One first notes that

$$\underline{\mathrm{Ext}}_Z^i(F, G) = \underline{\mathrm{Ext}}_{Z \cap \mathrm{Supp} F}^i(F, G).$$

(trivial, cf. Exposé ??). One first gives the proof for $Z = X$ hence $\mathrm{Supp} F \subset Y$. The functor

$$F \longmapsto \underline{\mathrm{Ext}}^n(F, G),$$

defined on the category of coherent $\mathcal{O}_X$ -Modules of support contained in Y , is left exact. By virtue of (?? ??), it is representable by

$$I = \varinjlim_k \underline{\mathrm{Ext}}^n(\mathcal{O}_X/\mathcal{I}^{k+1}, G),$$

where $\mathcal{I}$ is the ideal of definition of Y . Now, by (?? ??), one knows that:

$$\underline{H}_Y^n(G) \approx \varinjlim_k \underline{\mathrm{Ext}}^n(\mathcal{O}_X/\mathcal{I}^{k+1}, G).$$

Whence the conclusion if $Z = X$. Still by (?? ??) one knows that:

$$\underline{\mathrm{Ext}}_Z^n(F, G) \approx \varinjlim_k \underline{\mathrm{Ext}}^n(F/\mathcal{J}^{k+1}F, G),$$

where $\mathcal{J}$ is the ideal of definition of Z . The support of $F/\mathcal{J}^{k+1}F$ is contained in Y if $Z \cap \mathrm{Supp} F \subset Y$; by what we have just proved, one thus has:

$$\underline{\mathrm{Ext}}_Z^n(F, G) \approx \varinjlim_k \underline{\mathrm{Hom}}(F/\mathcal{J}^{k+1}F, \underline{H}_Y^n(G)).$$

It remains to show that the natural homomorphism:

$$\varinjlim_k \underline{\mathrm{Hom}}(F/\mathcal{J}^{k+1}F, \underline{H}_Y^n(G)) \longrightarrow \underline{\mathrm{Hom}}(F, \underline{H}_Y^n(G)),$$

is an isomorphism when $Z \cap \mathrm{Supp} F \subset Y$. Now X can be covered by affine noetherian open sets; one is thus reduced to the case where X is affine noetherian; then $F(X)$ is an $\mathcal{O}_X(X)$ -Module of finite type and $\mathrm{Supp} \underline{H}_Y^n(G) \subset Y$. Hence every homomorphism $u : F(X) \rightarrow \underline{H}_Y^n(G)(X)$ is annihilated by a power of $\mathcal{I}$ hence by a power of $\mathcal{J}$, qed.

PROPOSITION 1.2. *Let X be a locally noetherian prescheme, Y a closed subset of X , G a quasi-coherent $\mathcal{O}_X$ -Module and n an integer. Whatever be Z and S , closed subsets of X such that $Z \cap S = Y$, the following conditions are equivalent:*

- (i) $H_Y^i(G) = 0$ if $i < n$;
- (ii) there exists a coherent $\mathcal{O}_X$ -Module F , of support S , such that:

$$\underline{\text{Ext}}_Z^i(F, G) = 0 \text{ if } i < n;$$

- (iii) for every $\mathcal{O}_X$ -Module F coherent of support contained in S (i.e. $\text{Supp } F \cap Z = \text{Supp } F \cap Y$), one has:

$$\underline{\text{Ext}}_Z^i(F, G) = 0 \text{ if } i < n;$$

- (iv) for every coherent $\mathcal{O}_X$ -Module F , one has:

$$\underline{\text{Ext}}_Y^i(F, G) = 0 \text{ if } i < n.$$

Moreover, if they are satisfied, then for every coherent $\mathcal{O}_X$ -Module F and every closed subset Z' of X such that $Z' \cap \text{Supp } F = Y \cap \text{Supp } F$, one has isomorphisms:

$$\underline{\text{Ext}}_Z^n(F, G) \approx \underline{\text{Ext}}_Y^n(F, G) \approx \underline{\text{Hom}}(F, H_Y^n(G)).$$

PROOF. Let us argue by induction. The proposition is trivial for $n < 0$. Suppose it has been proved for $n < q$. If one of the conditions is satisfied for $n = q$, and for two subsets Z and S as stated, by the induction hypothesis one has, for every closed subset Z' of X and every coherent $\mathcal{O}_X$ -Module F such that $Z' \cap \text{Supp } F = Y \cap \text{Supp } F$, isomorphisms:

$$(1.1) \quad \underline{\text{Ext}}_{Z'}^{q-1}(F, G) \approx \underline{\text{Hom}}(F, H_Y^{q-1}(G)) \approx \underline{\text{Ext}}_Y^{q-1}(F, G).$$

Hence:

- (i) $\Rightarrow$ (iv), for one takes $Z' = Y$ in (1.1);
- (iv) $\Rightarrow$ (iii), for one takes $Z' = Z$ in (1.1);
- (iii) $\Rightarrow$ (ii), for one takes $F = \mathcal{O}_S$;
- (ii) $\Rightarrow$ (i), for one takes $Z' = Z$ in (1.1); whence $\underline{\text{Hom}}(F, H_Y^{q-1}(G)) = 0$; one then notes that:

$$\text{Supp } H_Y^{q-1}(G) \subset Y = Z \cap S \subset S = \text{Supp } F,$$

and one applies the following lemma:

LEMMA 1.3. *Let X be a prescheme, let P be a coherent $\mathcal{O}_X$ -Module, and let H be a quasi-coherent $\mathcal{O}_X$ -Module such that:*

$$\underline{\text{Hom}}(P, H) = 0 \text{ and } \text{Supp } P \supset \text{Supp } H.$$

Then $H = 0$.

It suffices to prove the lemma when X is affine, for the affine open sets form a basis of the topology of X and the hypotheses are preserved by restriction to an open set. Now, in this case, one is reduced to a problem on A -modules, where $X = \text{Spec}(A)$. One applies the formula (valid under the sole hypothesis that M is of finite type):

$$\text{Ass Hom}_A(P, H) = \text{Supp } P \cap \text{Ass } H;$$

One knows that $\text{Ass } H \subset \text{Supp } H \subset \text{Supp } P$ and that $\text{Ass Hom}_A(P, H) = \emptyset$; hence $\text{Ass } H = \emptyset$, hence $H = 0$.

To finish the proof of the proposition, it remains to note that (iv) allows one to apply 1.1. $\square$

COROLLARY 1.4. *Let G be a coherent and Cohen–Macaulay $\mathcal{O}_X$ -Module, let $n \in \mathbf{Z}$. The conditions of 1.2 are equivalent to:*

$$(v) \quad \text{codim}(Y \cap \text{Supp } G, \text{Supp } G) \geq n.$$

We first recall that an $\mathcal{O}_X$ -module is said to be Cohen–Macaulay if, for every $x \in X$, the fibre G_x is a Cohen–Macaulay $\mathcal{O}_{X,x}$ -module, i.e. one has for every $x \in S = \text{Supp } G$:

$$(1.2) \quad \text{prof } G_x = \dim G_x = \dim \mathcal{O}_{S,x}.$$

By Proposition ?? ??, condition (i) of 1.2 is equivalent to:

$$(1.3) \quad \text{prof}_Y G = \inf_{x \in Y} \text{prof } G_x \geq n,$$

hence also to:

$$\text{prof}_Y G = \inf_{x \in Y \cap S} \text{prof } G_x \geq n;$$

for the depth of a zero module is infinite.

Now, by definition:

$$\text{codim}(Y \cap S, S) = \inf_{x \in S \cap Y} \dim \mathcal{O}_{S,x},$$

whence the conclusion, applying the formula (1.2)).

We shall now prove a result allowing one to deduce the coherence conditions that we have in view from certain vanishing criteria.

LEMMA 1.5. *Let X be a locally noetherian prescheme. Let T^* be an exact contravariant ∂ -functor, defined on the category of coherent $\mathcal{O}_X$ -Modules, with values in the category of $\mathcal{O}_X$ -Modules. Let Y be a closed subset of X . Let $i \in \mathbf{Z}$. Suppose that, for every coherent $\mathcal{O}_X$ -Module of support contained in Y , $T^i F$ and $T^{i-1} F$ are coherent. Let F be a coherent $\mathcal{O}_X$ -Module. For $T^i F$ to be coherent, it is necessary and sufficient that $T^i F''$ be so, where one has set:*

$$F'' = F/\Gamma_Y(F).$$

Indeed, $F' = \Gamma_Y(F)$ is coherent because X is locally noetherian; the exact cohomology sequence of T^* then gives:

$$T^{i-1} F' \longrightarrow T^i F'' \longrightarrow T^i F \longrightarrow T^i F'$$

where the extreme terms are coherent, whence the conclusion.

LEMMA 1.6. *If F and G are coherent, and if $\text{Supp } F \subset Y$, $\underline{\text{Ext}}_Y^i(F, G)$ is coherent.*

Indeed, $\underline{\text{Ext}}_Y^i(F, G)$ is isomorphic to $\underline{\text{Ext}}^i(F, G)$; this is valid on any ringed space X , moreover: if Z is a closed subset which contains $Y \cap \text{Supp } F$, $\underline{\text{Ext}}_Z^i(F, G)$ is isomorphic to $\underline{\text{Ext}}_Y^i(F, G)$ (cf. Exposé ??).

PROPOSITION 1.7. *Suppose F and G are coherent and set $\text{Supp } F = S$, $S' = \overline{S \cap (X - Y)}$. Suppose that, for every $x \in Y \cap S'$, one has $\text{prof } G_x \geq n$, then $\underline{\text{Ext}}_Y^i(F, G)$ is coherent for $i < n$.*

Indeed, 1.6 allows one to apply 1.5 to $T^*(F) = \underline{\text{Ext}}_Y^*(F, G)$. Setting $F'' = F/\Gamma_Y(F)$, one sees that $\text{Supp } F'' = S'$. Now, by ?? ??, the hypothesis on the depth of G ensures the vanishing of $H_{Y \cap S'}^i(G)$ for $i < n$; by 1.2, one deduces from this the vanishing of $T^i F''$ for $i < n$, whence the conclusion thanks to 1.5.

2. Study for $i > n$

Let X be a locally noetherian *regular* prescheme, that is, one all of whose local rings are regular. Let Y be a closed subset of X . Let F and G be two coherent $\mathcal{O}_X$ -Modules. Set $S = \text{Supp } F$, $S' = \overline{S \cap (X - Y)}$. Set:

$$\begin{aligned} m &= \sup_{x \in Y \cap S} \dim \mathcal{O}_{X,x}, \\ n &= \sup_{x \in Y \cap S'} \dim \mathcal{O}_{X,x}; \end{aligned}$$

one has $n \leq m$.

PROPOSITION 2.1. *In the situation described above, one has:*

- (1) $\underline{\text{Ext}}_Y^i(F, G) = 0$ if $i > m$,
- (2) $\underline{\text{Ext}}_Y^i(F, G)$ is coherent if $i > n$.

One first notes that $\underline{\text{Ext}}_Y^i(F, G)$ is coherent for every i when $\text{Supp } F \subset Y$. Moreover, setting as above $F'' = F/\Gamma_Y(F)$, one sees that $\text{Supp } F'' = S'$, hence (2) follows from (1) and from 1.3.

To prove (1), one first notes that

$$\underline{\text{Ext}}_Y^i(F, G) \approx \varinjlim_k \underline{\text{Ext}}^i(F/\mathcal{J}^k F, G),$$

where $\mathcal{J}$ is the ideal of definition of Y . Moreover, it follows from theorem 4.2.2. of (A. Grothendieck, “Sur quelques points d’algèbre homologique”, *Tôhoku Mathematical Journal* **9** (1957), p. 119–221.) that the $\underline{\text{Ext}}$ commute with the formation of fibres, at least when X is a locally noetherian prescheme and when the first argument is coherent; as the same holds for inductive limits, one finds isomorphisms:

$$(\underline{\text{Ext}}_Y^i(F, G))_x \approx \varinjlim_k \text{Ext}_{\mathcal{O}_{X,x}}((F/\mathcal{J}^k F)_x, G_x)$$

for every $x \in X$. As $\text{Supp } \underline{\text{Ext}}_Y^i(F, G) \subset S \cap Y$, it suffices, in order to conclude, to note that $x \in Y \cap S$ implies $\dim \mathcal{O}_{X,x} \leq m$, hence:

$$\text{Ext}_{\mathcal{O}_{X,x}}^i((F/\mathcal{J}^k F)_x, G_x) = 0 \text{ if } i > m,$$

for the global cohomological dimension of a regular local ring is equal to its dimension¹.

Let X be a locally noetherian prescheme; for every subset P of X , set:

$$D(P) = \{\dim \mathcal{O}_{X,p} \mid p \in P\}.$$

LEMMA 2.2. *If P is the underlying space of a connected subprescheme of A , $D(P)$ is an interval.*

Indeed, let a and b belong to $D(P)$, corresponding to points p and q of P . Let us show that there exists a sequence of points of P : $(p = p_1, \dots, p_n = q)$ such that, for $1 \leq i < n$, one has $|\dim \mathcal{O}_{X,p_i} - \dim \mathcal{O}_{X,p_{i+1}}| = 1$; it will follow that $D(P)$ contains the interval $[p, q]$. For this, one notes that p and q can be joined by a sequence of irreducible components of P , such that two successive components meet. One is reduced to the case where p is the generic point of an irreducible component Q of P , and where $q \in Q$ and hence $q \supset p$, as ideals of $\mathcal{O}_q$, where it is trivial from the definition of dimension.

PROPOSITION 2.3. *Let X be a locally noetherian regular prescheme, Y a closed subset of X and F a coherent $\mathcal{O}_X$ -Module. Let $P = Y \cap \text{Supp } F \cap (X - Y)$. Let $n \in \mathbf{Z}$, and suppose that $n \notin D(P)$. Then $\underline{\text{Ext}}_Y^n(F, \mathcal{O}_X)$ is coherent.*

The conclusion is local and the hypotheses are preserved by restriction to an open set. Now P is closed hence locally noetherian, hence locally connected; one may thus assume X affine and noetherian, and P connected. Set $D(P) = [a, b]$, which is legitimate by the preceding lemma; if $n > b$, one concludes by 2.1; if $n < a$, one has $n < \dim \mathcal{O}_{X,x} = \text{prof } \mathcal{O}_{X,x}$ for every $x \in P$, and one concludes by 1.7.

¹Cf. EGA 0_{IV} 17.3.1